Driver Installation

Windows System

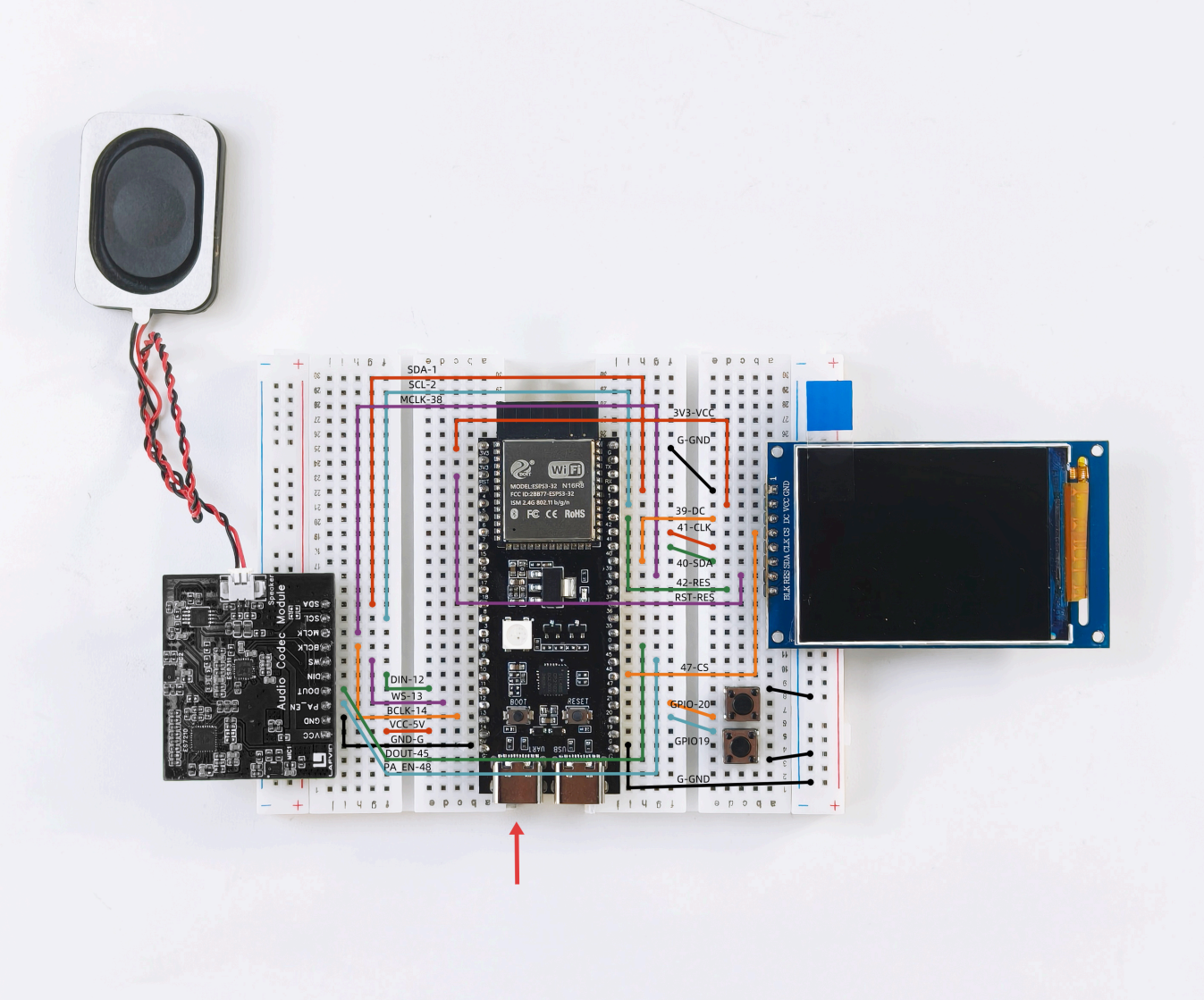
Note

On Windows 11, the CP210x driver may install automatically after you connect the board to the computer. If no serial port appears in Device Manager, install the Silicon Labs CP210x driver manually.

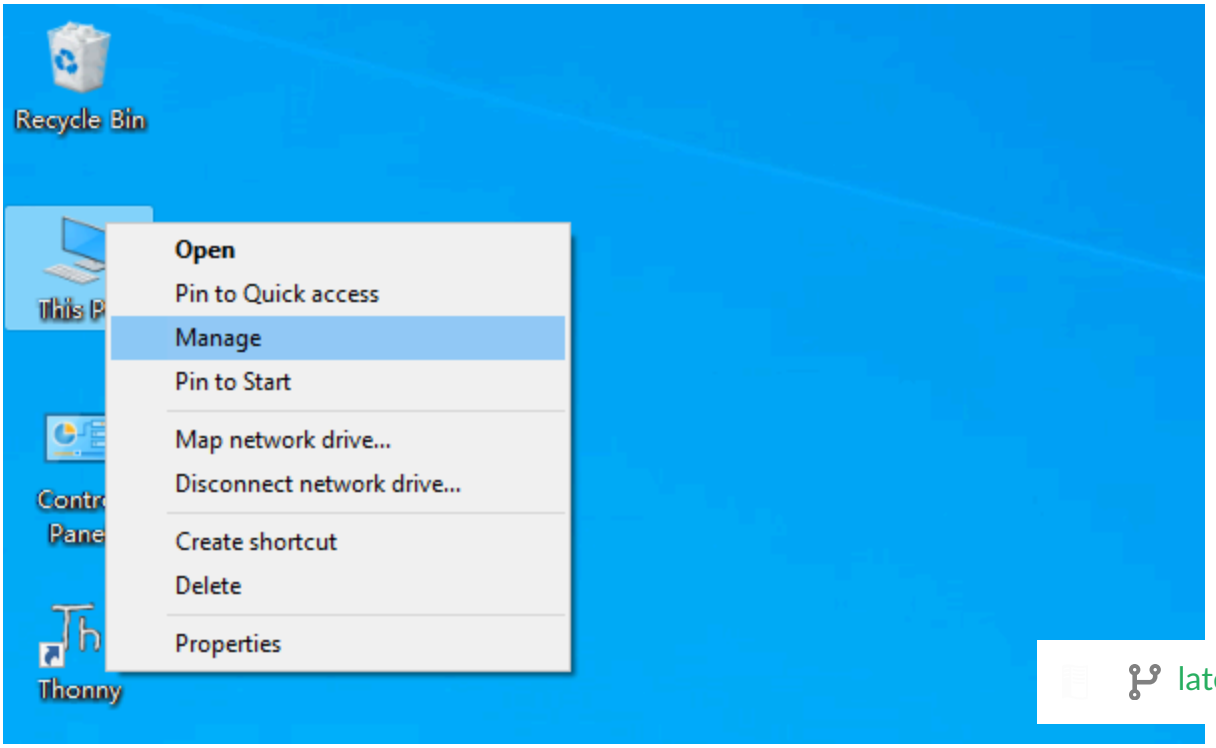
Check If CP210X Is Already Installed

ESP32-S3 uses CP210X for code downloading. Before using it, we need to install the CP210X driver on our computer.

1. Connect your computer and ESP32-S3 with a USB cable (connect to the port indicated by the red arrow)



2. Go to your computer's main interface, select "This PC" and right-click to select "Manage".

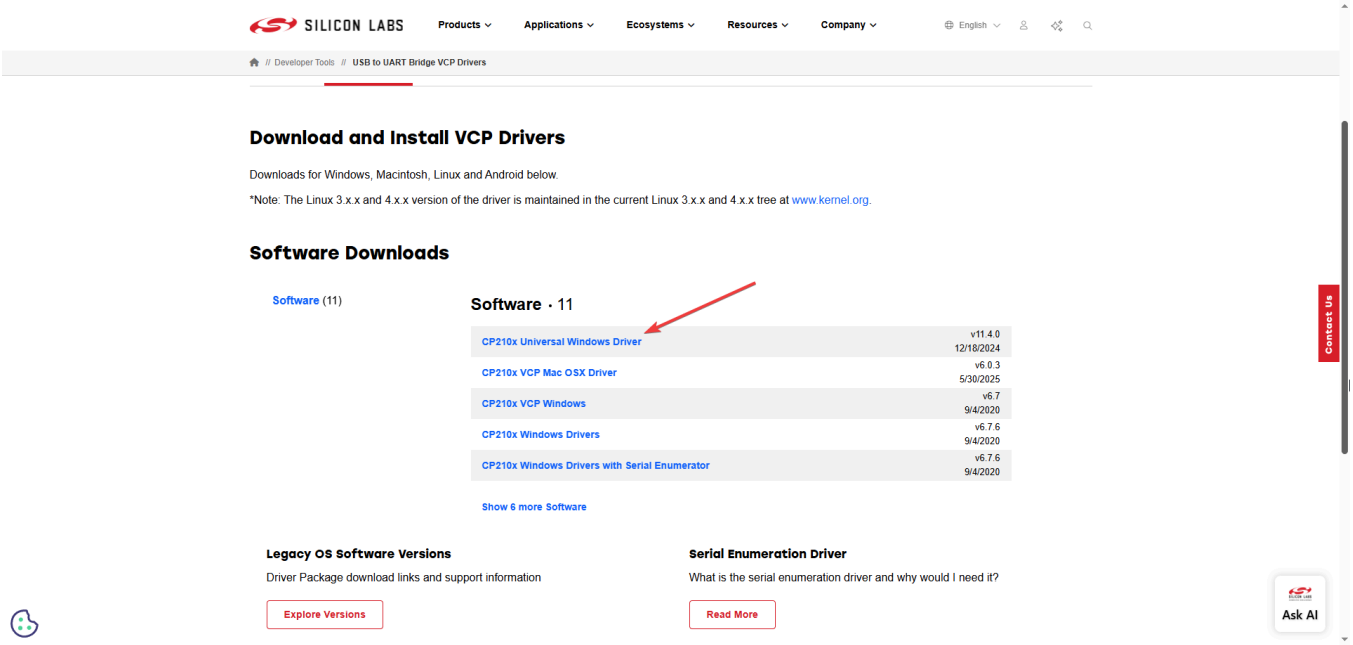


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3. Click “Device Manager”. If your computer has already installed CP210X, you will see “USB-Enhances-SERIAL CP210X (COMx)”. You can then proceed to the next step.

Installing CP210X

1. First, download the CP210X driver. Click HERE to download the appropriate driver for your operating system.



If you don’t want to download the installation package from the website, you can open “LAFVIN-AI-Chatbot/CP210X”, as we have already prepared the installation package.

You can install it following the video below

Windows 11 Driver Troubleshooting

If the CP210x driver cannot be installed or the serial port does not appear:

1. Make sure you are using a USB Type-C cable that supports data transfer.
2. Connect the cable to the ESP32-S3 `UART` port, not the `USB` port.
3. Open Device Manager and check both `Ports (COM & LPT)` and `Other devices`.
4. If you see an unknown device or a CP210x device with a warning icon, right-click it and choose `Update driver`.
5. Use the official Silicon Labs driver whenever possible. Avoid installing random driver packages from unofficial sources.
6. After installing the driver, unplug and reconnect the USB cable, or restart Windows if the COM port still does not appear.
7. If it still fails, contact support with your Windows version, a screenshot of D and the driver package you tried to install.

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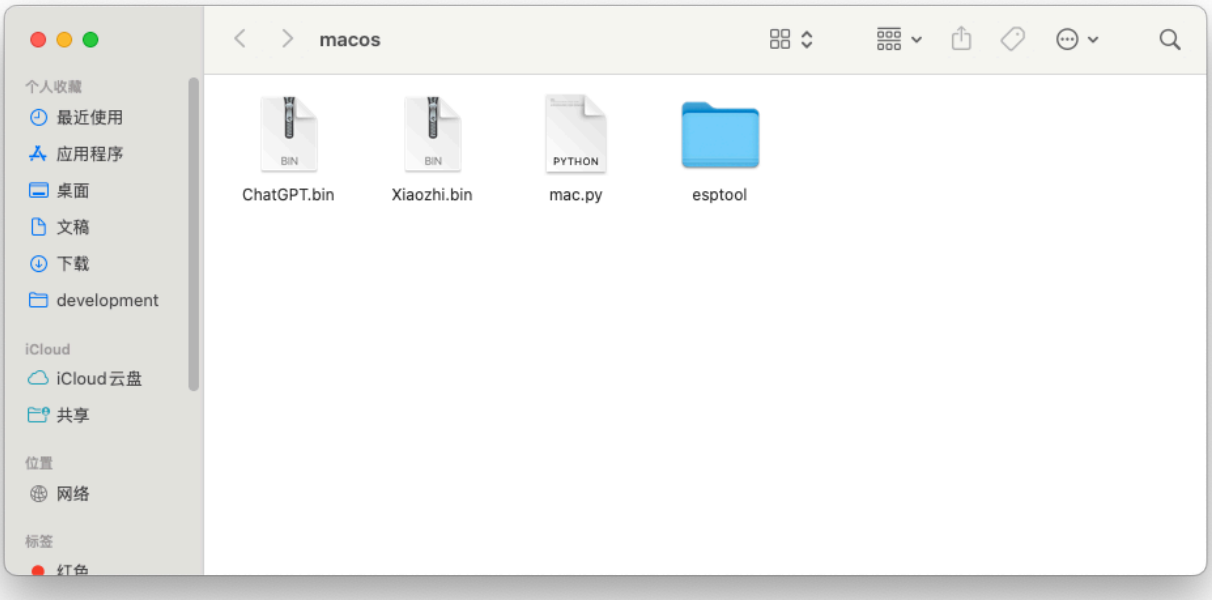
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macOS Firmware Upload (Alternative)

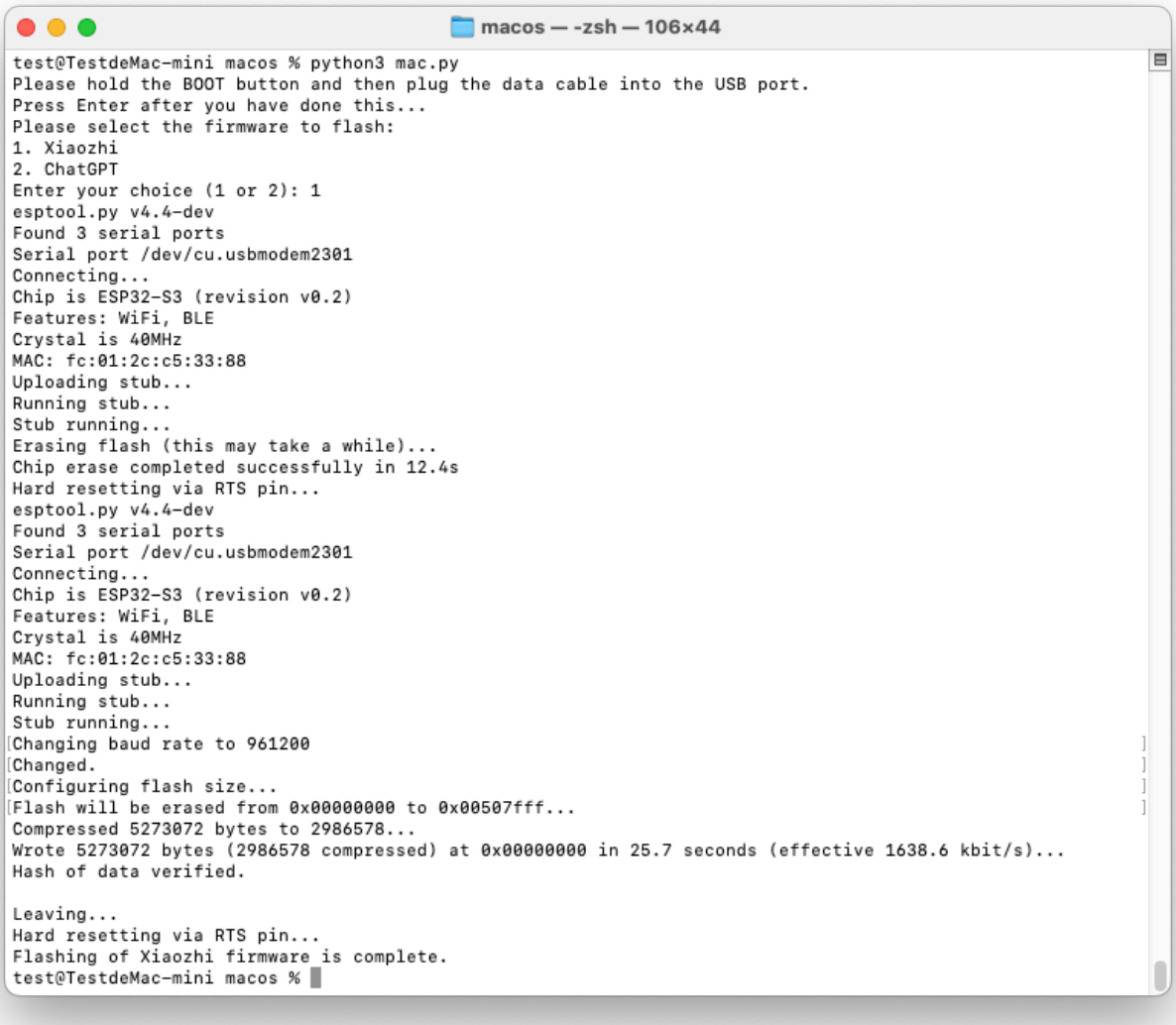
For most users running macOS Catalina (10.15) or later, the CP210x driver is built into the system, so no separate driver installation is required.

> **Note**
>
> We recommend using the Online Flasher first. The local macOS flashing method below is an alternative option when you cannot use online flashing.

1. First, find the macOS folder in the GitHub repository
2. We have prepared the flashing tools and files for macOS in this folder



3. Open Terminal, then run `python3 mac.py`. The following interface will appear.
4. Press and hold the `BOOT` button on the ESP32S3, connect the USB Type-C cable to the `USB` port, then press `Enter` and select the AI firmware you want to flash.



5. Click `Confirm`, and the program will automatically complete the flashing process.
6. After flashing is complete, press the `RESET` button on the ESP32S3 and wait for the device to restart.
7. After restarting, follow xiaozhi_conf or chatgpt_conf for network and API configuration, then add the device in the backend.